



RIVER VALLEY HIGH SCHOOL YEAR 6 PRELIMINARY EXAMINATION II

H2 BIOLOGY 9648

PAPER 1
25 SEP 2014

1 HOUR 15 MIN

NAME _____

CLASS 6 ()

INDEX NO. _____

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Write your name, class and index number on the Answer Sheet in the spaces provided unless this has been done for you.

DO **NOT** WRITE IN ANY BARCODES.

There are **forty** questions in this paper. Answer **all** questions. For each question, there are four possible answers, **A**, **B**, **C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

Any rough working should be done in this booklet.

The use of an approved scientific calculator is expected, where appropriate.

This Question Paper consists of **27** printed pages.

For each question, there are four possible answers, **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

1. The diagram shows a section of a cell. Which of the following correctly matches the structure of the labelled component to its function?

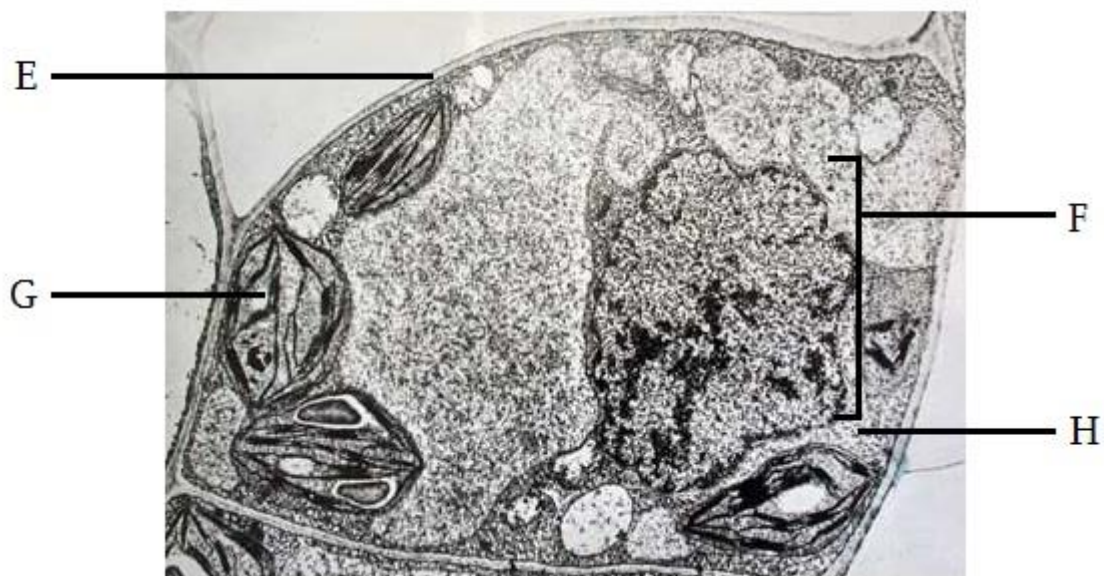


Image adapted from <http://umanitoba.ca/Biology/BIOL1020/lab3/images/PlantCell-Micrograph1.jpg>

	Component	Structure	Function
1	E	Extensive hydrogen bonds between tropocollagen molecules	High tensile strength to maintain cell shape
2	F	Selectively permeable due to its phospholipid bilayer	Regulate entry and exit of ribosomal proteins
3	G	ATP synthetase embedded in inner membrane	Synthesises ATP for Krebs cycle
4	H	Presence of glycolytic enzymes	Oxidises glucose to pyruvate

- A** 2 only
B 2 and 4 only
C 2, 3 and 4
D All of the above

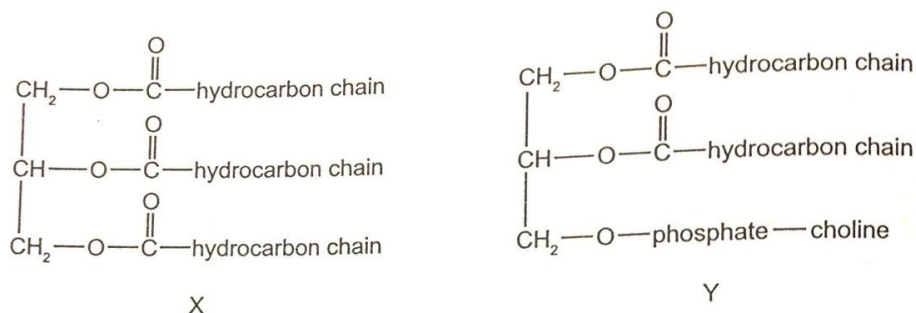
2. Three pure substances are analysed. The table shows the elements that each contains.

	C	H	O	N	P	S
X	+	+	+	+	-	-
Y	+	+	+	-	-	-
Z	+	+	+	+	-	+

Which combination could be correct?

	X	Y	Z
A	adenine	maltose	triglyceride
B	adenine	triglyceride	amino acid
C	amino acid	triglyceride	maltose
D	triglyceride	maltose	amino acid

3. The diagram shows the structure of two lipid molecules.

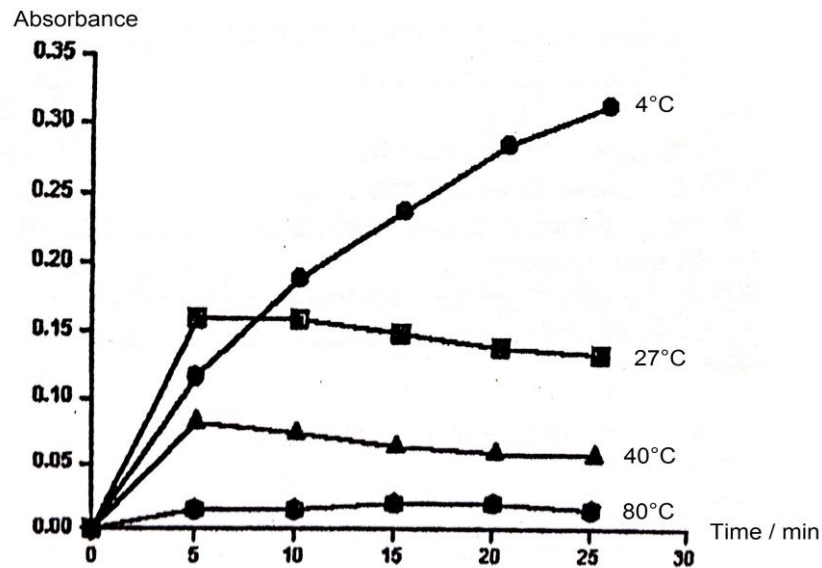


Which statements correctly describe the two lipid molecules?

- Both molecules are esters of glycerol.
- Both the molecules are made by condensation reactions.
- The hydrocarbon chains of molecule X are always from saturated fatty acids.
- The hydrocarbon chains of the molecule Y may be from saturated or unsaturated fatty acids.
- The hydrocarbon chains of molecule Y are always same length.

- A 1 and 2 only
 B 1 and 3 only
 C 1, 2 and 4
 D 2, 3 and 5

4. The diagram shows the activity of an enzyme X found in bacteria species Z. The y-axis shows the absorbance caused by the presence of product.



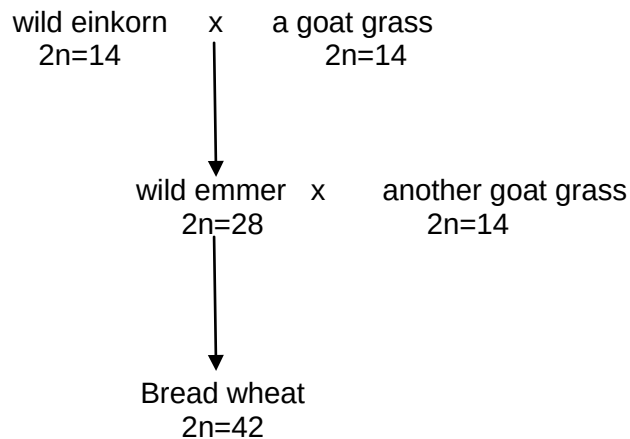
Which of the following is least likely to be true?

- A Bacteria species Z is most likely not pathogenic to humans.
 - B Bacteria species Z thrives in conditions below 4°C.
 - C Increasing temperatures denature enzyme X.
 - D Active site for enzyme X is saturated at 30 min.
5. A toxic chemical causes malfunction of the centrioles in animal cells.

Which process in mitosis is likely to be directly affected by the chemical?

- 1 migration of the chromosomes
 - 2 replication of centromeres
 - 3 chromosome shortening
 - 4 pairing of chromatids
 - 5 alignment of chromosomes
- A 1 and 5 only
 - B 2 and 5 only
 - C 1, 3 and 5
 - D 2, 3 and 5

6. Bread wheat is the result of multiple crosses between ancient wheat species and goat grasses over the last ten thousand years.



Why does this type of chromosome change not normally happen?

- A Diploid cells are formed by mitosis.
 - B Diploid gametes undergo a reduction division.
 - C Meiosis halves the diploid number of chromosomes.
 - D Meiosis results in diploid gametes.
7. In an attempt to synthesise DNA molecules *in vitro*, a student isolated and purified various molecules needed for DNA replication. She added some DNA to the mixture, and replication occurred. However, the DNA molecules formed were defective. Each molecule consists of a normal DNA strand paired with numerous segments of DNA, each about hundreds of nucleotides long. What might she have left out in the mixture?
- A DNA primer
 - B RNA primer
 - C DNA ligase
 - D DNA polymerase

8. A pure sample of DNA was obtained and transcribed *in vitro*. The mRNA obtained was purified and analysed for its base composition. The table shows the results of analysis for the said sample of DNA and the purified mRNA.

	Base composition / %				
	A	T	G	C	U
DNA Strand 1	19.1	23.9	26.0	31.0	0
DNA Strand 2	24.2	19.3	30.8	25.7	0
mRNA	19.0	0	25.9	30.8	24.3

Which of the following is a valid conclusion from the results shown?

- A DNA Strand 1 acts as a template for the synthesis of mRNA.
 - B DNA Strand 2 acts as a template for the synthesis of mRNA.
 - C Neither DNA Strand 1 nor DNA Strand 2 is the template for the synthesis of mRNA.
 - D None of the above.
9. The following statements describe various steps in translation.
- 1 Large ribosomal subunit binds to mRNA.
 - 2 Small ribosomal subunit binds to mRNA.
 - 3 Anticodon of activated tRNA base pairs with the codon at the A site.
 - 4 Anticodon of activated tRNA base pairs with its codon on mRNA.

Which of the following statements describe the initiation phase?

- A 1 and 2 only
- B 1, 2 and 3
- C 1, 2 and 4
- D All of the above

10. Which of the following mutations may result in complete loss of expression of the peptide sequence of an affected gene?

- 1 Substitution
- 2 Inversion
- 3 Deletion
- 4 Insertion

- A** 1 only
- B** 3 and 4 only
- C** 1, 3 and 4
- D** All of the above

11. One method of limiting damage to cells from viral diseases uses an antisense-RNA for a protein formed late in the virus cycle.

Which stage(s) of a retrovirus life cycle would be inhibited by this antisense RNA?

- 1 activity of restriction endonucleases
- 2 assembly of coat proteins
- 3 formation of cDNA
- 4 activity of reverse transcriptase
- 5 activity of protease

- A** 2 only
- B** 3 only
- C** 3 and 4 only
- D** 1, 4 and 5

12. Which of the following statements about *lac* operon is/are true?

- 1 *lac* operon is switched off when glucose is no longer available and repressor becomes active as glucose leaves it.
- 2 *lac* operon is switched off when glucose is available and the repressor becomes active as lactose binds to it.
- 3 *lac* operon is switched on when RNA polymerase binds to the promoter once the repressor is inactive.
- 4 *lac* operon is switched on when the regulator gene is switched off due to the presence of lactose.

- A** 3 only
B 1 and 3 only
C 2 and 4 only
D 1, 3 and 4

13. Lactose is a disaccharide that can be used by *E.coli* as source of energy instead of glucose. Genes involved in lactose metabolism are called the *lacZ*, *lacY* and *lacA*. In wild type bacteria, *lacZ*, *lacY* and *lacA* expression rises dramatically when the lactose concentration rises and glucose concentration falls.

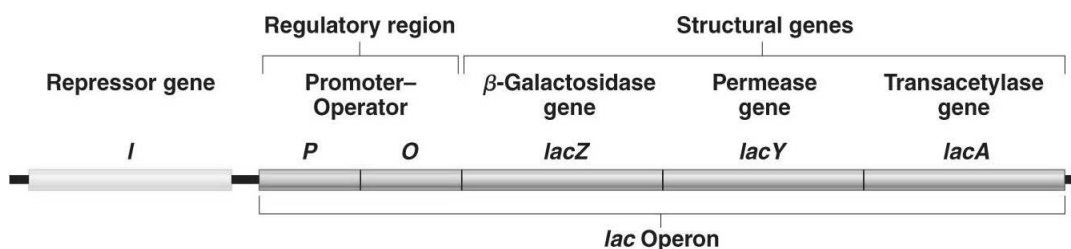


Image adapted from Pearson Education Inc, 2009

A mutant strain of *E.coli* abnormal in its use of lactose was identified. The tables show the expression of each protein by *E.coli* when grown in the absence and presence of lactose. Assume lactose as the only carbon source.

(✓ = gene is expressed; × = gene is not expressed)

	Without lactose		
	<i>lacZ</i>	<i>lacY</i>	<i>lacA</i>
Wild type	×	×	×
Mutant strain	×	×	×

	With lactose		
	<i>lacZ</i>	<i>lacY</i>	<i>lacA</i>
Wild type	✓	✓	✓
Mutant strain	✓	×	×

Which of the following best accounts for the observed data?

- A A base-pair deletion in *lacY* coding sequence
- B A base-pair deletion in *lacA* coding sequence
- C Presence of premature stop codon in *lacY* gene
- D Presence of premature stop codon in *lacZ* gene

14. The diagram shows the folding of eukaryotic DNA.

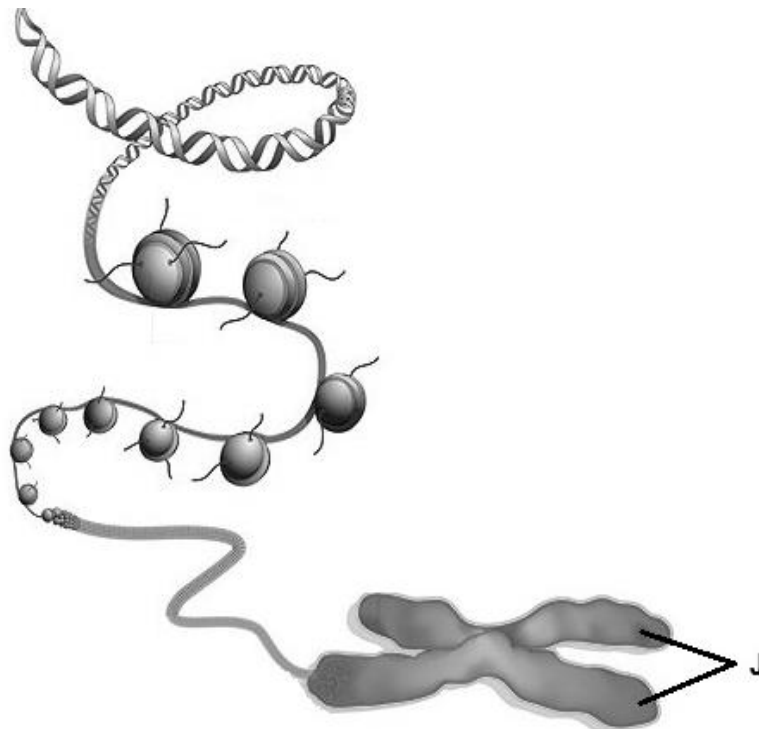


Image adapted from <http://www.flickr.com/photos/moonmoondeb/4441861091/in/pool-1309029@N95>

In which of the following events would DNA be folded to same level of condensation as the structure labelled **J**.

- 1 Anaphase II of meiosis
 - 2 Metaphase of mitosis
 - 3 S phase of cell cycle
 - 4 Late telophase of mitosis
- A** 2 only
B 1 and 2 only
C 1, 2 and 3
D 1, 2 and 4

- 15.** Jett wanted to determine the function of protein **X**. He cloned gene **X** into plasmid **1**. He then set up another plasmid comprising a binding site for the product of gene **X** and a reporter gene. A reporter gene codes for the synthesis of a product that is easily measured. The two plasmids were then introduced into the nucleus of a suitable host cell.

He noted that there was an increase in the number of reporter gene transcripts. Which of the following is a valid conclusion?

- A** Gene **X** is an enhancer sequence.
 - B** Gene **X** is the promoter sequence.
 - C** Product **X** is RNA polymerase.
 - D** Product **X** is a transcription factor.
- 16.** Which of the following is a method of post-transcriptional control in cells?
- 1 Down-regulation of eukaryotic Initiation Factor 2 (eIF2) in Alzheimer's disease.
 - 2 Selective degradation of mRNA by cytoplasmic exonucleases in *Bacillus subtilis*.
 - 3 Prevention of ribosome attachment at 5'UTR by Iron Regulatory Protein (IRP).
 - 4 Conversion of pro-insulin to functional insulin via cutting of C chain.
- A** 2 only
 - B** 1 and 3 only
 - C** 1 and 4 only
 - D** None of the above
- 17.** Which of the following statements about proto-oncogene is true?
- 1 Proto-oncogene encodes proteins which stimulate the cell cycle.
 - 2 Proto-oncogenes and tumour suppressor genes work the same way.
 - 3 When the transcription of proto-oncogene becomes abnormal, they can become oncogenes.
 - 4 Only one copy of proto-oncogene needs to be mutated to synthesise a hyperactive protein product.
- A** 1 only
 - B** 4 only
 - C** 1 and 3 only
 - D** 1, 3 and 4

18. In rats, the allele of a gene for 'mottled' coat (M) and the recessive allele (m) for 'normal' coat are sex linked. The allele of a gene for 'long' whiskers (W) and the recessive allele (w) for 'short' whiskers are autosomal.

A male rat with a normal coat and short whiskers was mated on several occasions to the same female. The offspring showed the following phenotypes in equal proportions.

- Mottled females and males with long whiskers
- Mottle females and males with short whiskers
- Normal females and males with long whiskers
- Normal females and males with short whiskers

What are the genotypes of the parents?

- A X^MYww and X^MX^MWW
- B X^mYww and X^MX^MWw
- C X^mYww and X^MX^mWW
- D X^mYww and X^MX^mWw
19. Thalassemia is a hereditary disease of the blood resulting in anaemia. Severe anaemia (thalassemia major) is found in homozygotes (T^MT^M) and a milder form (thalassemia minor) is found in heterozygotes (T^MT^m).

Assuming that all offspring with thalassemia major die before adulthood, what proportion of the adult generation from marriages in which both partners have thalassemia minor would have anaemia?

- A 0.25
- B 0.33
- C 0.50
- D 0.66

20. In Mendel's experiment, he used green pea pods from the F₁ generation and obtained 428 green pea pods and 152 yellow pea pods in F₂ generation. There was no significant difference, at a 95% confidence level, between the observed and the expected results.

Table of chi-squared values					
Degree of freedom	Probability P (%)				
	95	80	50	20	5
1	0.00393	0.0642	0.455	1.642	3.841
2	0.103	0.446	1.386	3.219	5.991
3	0.352	1.005	2.366	4.642	7.816
4	0.711	1.649	3.357	5.989	9.48

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

O= observed result
E= expected result
v= n-1

Using the equation and the table of chi-squared values above, which value would be close to the calculated chi-squared value for this experiment?

- A 0.388
- B 3.34
- C 3.88
- D 9.52

21. Two genes involved in coat colour of goats are located on different chromosomes.

The colour gene C causes the hairs to have uniform colour and has three alleles.

- C^{DB} giving dark brown hairs
- C^B giving black hairs
- C^{MB} giving medium brown hairs.

A dominant allele of the agouti gene (A^G) causes the development of white hairs between the coloured hairs giving the coat a shaded appearance.

The table shows the results of crosses between a male goat and two female goats.

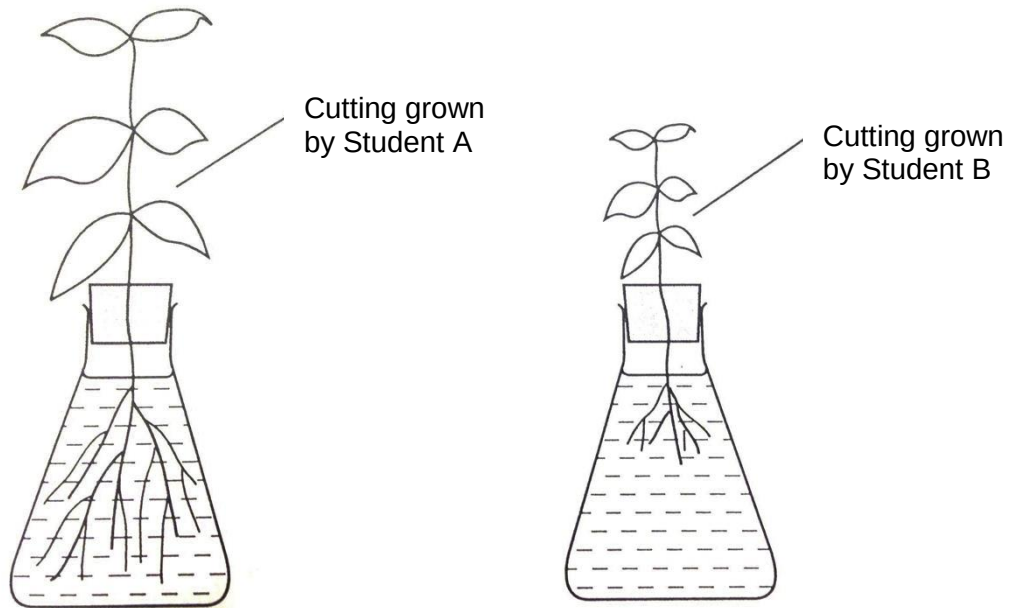
parents	Offspring
black agouti male x uniformly dark brown female	50% uniform, 50% agouti; 50% dark brown, 50% black
black agouti male x medium brown agouti female	all agouti; 50% black, 50% medium brown

Which row shows the possible genotypes of the male and female goats?

	black agouti male	uniformly dark brown female	medium brown agouti female
A	$C^B C^{MB} A^G A^G$	$C^B C^{DB} A^G A^g$	$C^{MB} C^{MB} A^G A^G$
B	$C^B C^{MB} A^G A^g$	$C^B C^{DB} A^g A^g$	$C^B C^{MB} A^G A^G$
C	$C^B C^{DB} A^G A^G$	$C^{DB} C^{MB} A^G A^g$	$C^B C^{DB} A^G A^g$
D	$C^B C^{MB} A^G A^g$	$C^B C^{DB} A^g A^g$	$C^{MB} C^{MB} A^G A^G$

- 22.** Student A was tasked to look after a plant, while Student B looked after another plant of the same species. Each student followed the same instructions to set up the apparatus to take cuttings from their plant and grow the cuttings next to one another.

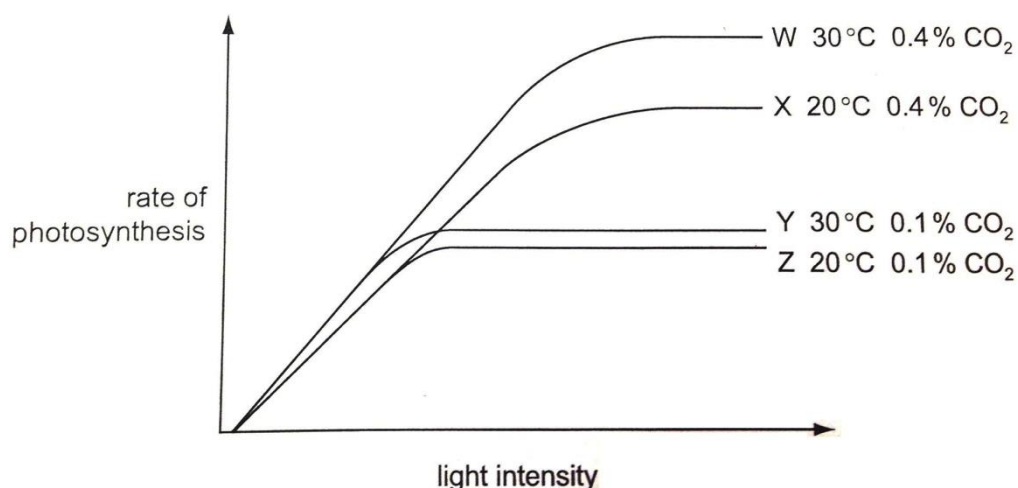
The diagrams show the results after one week.



Which factors could have caused the different appearance of the two cuttings?

- 1 phenotypic variation due to environment
 - 2 genetically different cuttings
 - 3 mutation due to environment
 - 4 genotypic variation due to environment
- A** 1 and 2 only
B 2 and 3 only
C 2 and 4 only
D 1, 3 and 4

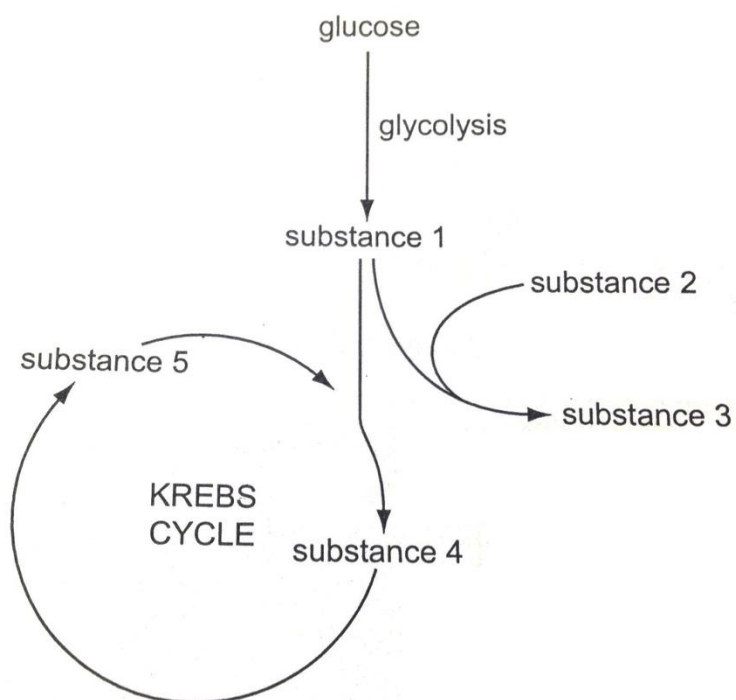
23. The diagram shows the results of an investigation into the effect of changing light intensity on the rate of photosynthesis at two different carbon dioxide concentrations and two different temperatures.



Which of the following is true for curve X and which curve supports this?

	Limiting factor	Evidence
A	carbon dioxide	curve Y as a decrease in carbon dioxide concentration decreases rate
B	carbon dioxide	curve Z as rate becomes constant at lower light intensities
C	temperature	curve W as an increase in temperature increases rate
D	temperature	curve Z as rate becomes constant at lower light intensities

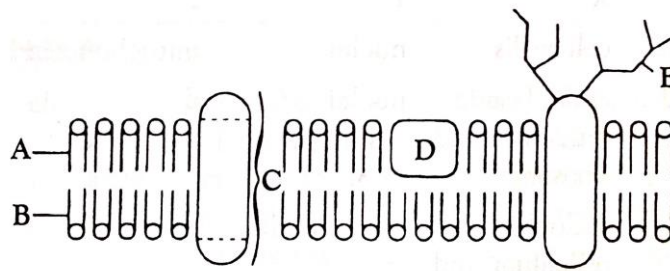
24. The diagram represents some of the stages of anaerobic and aerobic respiration.



Which row shows the identity of substances 1, 2, 3, 4 and 5?

	1	2	3	4	5
A	Acetyl coA	NAD	ethanol	pyruvate	6 carbon compound
B	Acetyl coA	reduced NAD	lactate	6 carbon compound	4 carbon compound
C	pyruvate	NAD	ethanol	acetyl coA	6 carbon compound
D	pyruvate	reduced NAD	lactate	6 carbon compound	4 carbon compound

25. The diagram represents a model of the structure of the biological membrane.



Which of the following statements are correct?

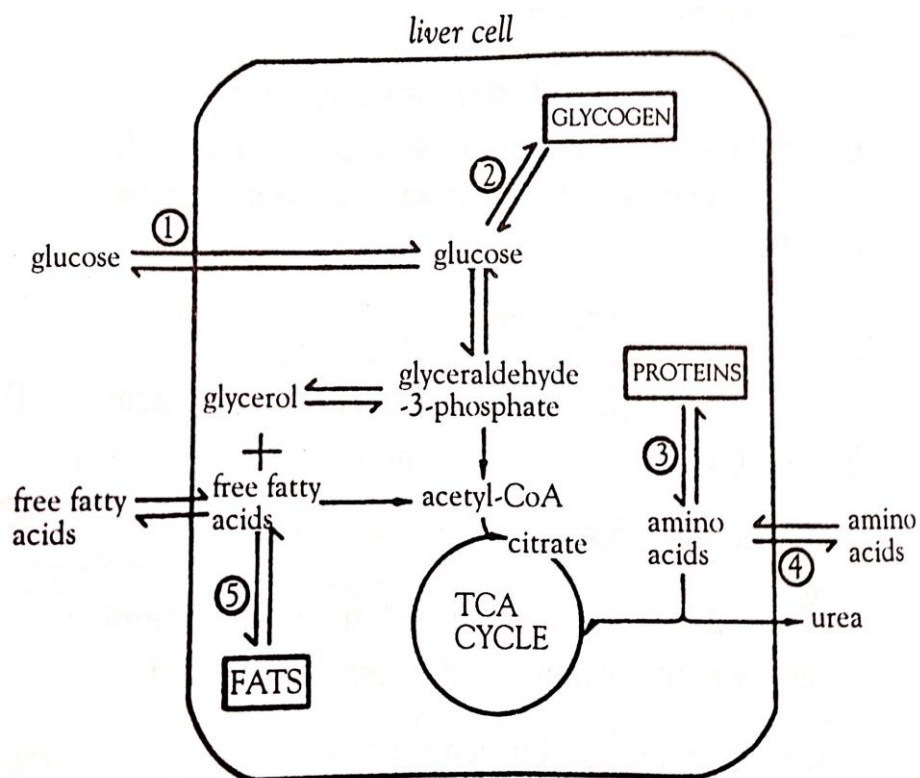
- 1 Structures C and E would produce amino acids after hydrolysis.
- 2 Polar molecules require structure D to enter the cell.
- 3 Structure C may be involved in cell signalling.
- 4 Structures A and B give rise to selectively permeability.

- A** 2 and 4 only
B 3 and 4 only
C 1, 2 and 4
D 2, 3 and 4

26. Potassium cyanide is known to interfere with the formation of ATP in cell metabolism. If the use of potassium cyanide resulted in the accumulation of a solute in a cell, which of the following is a valid conclusion?

- A** The solute normally enters the cell by active transport.
B The solute normally enters the cell by diffusion.
C The solute normally enters the cell by pinocytosis.
D The cell is plasmolysed.

27. The diagram shows some biochemical pathways in a liver cell.



At which numbered points would the hormone insulin accelerate the pathway in the directions indicated?

- A 1, 2 and 3
- B 1, 2 and 5
- C 1, 3 and 4
- D 3, 4 and 5

28. The following sentences describe the sequence of events in the Jak-STAT cell signalling pathway.

- 1 α -interferon binds to the ligand binding site on the transmembrane α -interferon receptor.
- 2 Receptor subunits, Tyk2 and Jak1, associates closely with one another.
- 3 Activated receptor binds cytoplasmic relay proteins at their SH domains.
- 4 Cell proliferation is stimulated.

Which of the following pairs, when taken together, suggest the presence of a relay system?

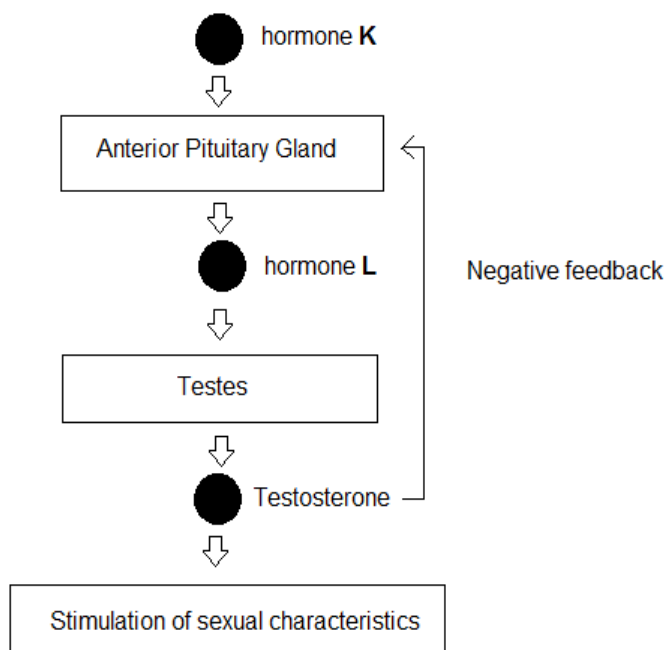
- A** 1 and 2
- B** 1 and 3
- C** 2 and 3
- D** 2 and 4

29. Which of the following would stop impulse transmission across a synapse?

- 1 A mutation of the voltage-gated calcium ion channels.
- 2 Enzyme cholinesterase inhibited by nerve gas.
- 3 Increased fusion of neurotransmitter vesicles with post-synaptic membrane.
- 4 Accumulation of acetylcholine vesicles in the pre-synaptic knob.

- A** 4 only
- B** 1 and 2 only
- C** 1 and 3 only
- D** 1 and 4 only

30. The production of testosterone is regulated by the anterior pituitary gland. Upon stimulation by hormone **K**, the gland releases hormone **L** which causes the testes to produce testosterone. The elevated level of testosterone exerts a negative feedback on the anterior pituitary gland.



If an individual consumes large amount of steroids containing testosterone, which of the following would be observed?

	Production of hormone K	Production of hormone L	Production of testosterone	Stimulation of sexual characteristics
A	Decrease	Decrease	Decrease	No change
B	Decrease	Decrease	No change	Increase
C	No change	Decrease	Decrease	Increase
D	No change	Decrease	Decrease	No change

- 31.** Bacteria in the genus *Wolbachia* infect many butterfly species. They are passed from one generation to the next in eggs, but not in sperm, and they selectively kill developing male embryos.

In Samoa in the 1960s, the proportion of male blue moon butterflies fell to less than 1% of the population. However, by 2006, the proportion of males was almost 50% of the population.

Resistance to *Wolbachia* is the result of the dominant allele of a suppressor gene.

Which statements correctly describe the evolution of resistance to *Wolbachia* in the blue moon butterfly population?

- 1 *Wolbachia* acts as a selective agent.
 - 2 The selective killing of male embryos is an example of artificial selection.
 - 3 When infected with *Wolbachia*, male embryos that are homozygous for the recessive allele of the suppressor gene die.
 - 4 All male embryos that carry the dominant allele of the suppressor gene pass that allele to their offspring.
 - 5 The frequency of the dominant allele of the suppressor gene rises in the butterfly population.
- A** 1 and 4 only
B 2 and 3 only
C 1, 3 and 5
D 2, 4 and 5

32. The α , β and γ globin chains of human, chimpanzee, gorilla and gibbon haemoglobin were analysed. The number of differences in the amino acid sequences in comparison with the human haemoglobin is shown in the table.

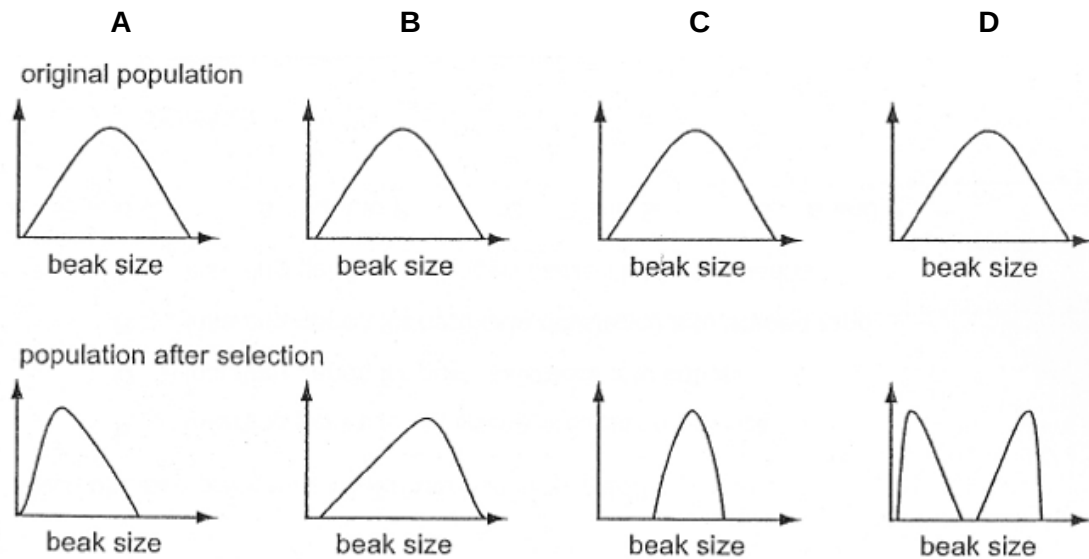
species	number of differences in amino acid sequence from human molecules		
	α globin	β globin	γ globin
chimpanzee	0	0	1
gorilla	1	1	1
gibbon	3	3	2

Which of the following(s) can be concluded from the information given in the table?

- 1 Humans and gorillas shared a common ancestor more recently than humans and gibbons.
 - 2 Humans and gibbons shared a common ancestor more recently than humans and chimpanzees.
 - 3 Humans evolved from chimpanzees.
 - 4 Humans and gorillas do not share a common ancestor.
- A** 1 only
B 2 only
C 1 and 3 only
D 3 and 4 only

33. One species of finch on an island shows variation in beak size. Birds with larger beaks can eat larger seeds. Following a dry period, fewer small seeds are available, with larger seeds and nuts being more plentiful.

Which graph shows the effect of natural selection on beak size of these birds following a dry period?



34. Which of the following is a reason why the construct of cDNA library is preferred over that of genomic DNA library?
- A Construct of cDNA library does not require the use of bacteria host cells.
 - B cDNA constructs provides information about alternative splicing of mRNA.
 - C cDNA library for a particular cell type is the same for all individuals of a species.
 - D Regulatory sequences such as enhancers and silencers can only be found in cDNA constructs.

- [illegible]

A Adult 1 and Adult 2
B Adult 1 and Adult 3
C Adult 2 and Adult 4
D Adult 2 and Adult 3

- Which of the following explains this choice of DNA polymerase?

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- 37.** Which of the following poses the greatest limitation to the Human Genome Project in the area of disease analysis?
- A** It is not possible to store the genetic information of all individuals.
 - B** There are too few model organisms for comparative genome studies.
 - C** Environment factors and gene interaction in diseases are not taken in to account.
 - D** There is too large a genetic difference between individuals to determine the loci of genes of interest.
- 38.** Which of the following is/ are the reasons that make embryonic stem cells particularly useful in medical research?
- 1 They can be fused together to form a zygote.
 - 2 They have the characteristics of a multipotent stem cell.
 - 3 They will continue to divide indefinitely.
 - 4 They can be stimulated by chemical signals to express particular genes.
 - 5 They can only differentiate into a particular tissue.
- A** 3 only
 - B** 2 and 3 only
 - C** 2, 3 and 4
 - D** 3, 4 and 5
- 39.** A patient suffering from cystic fibrosis underwent gene therapy using a viral gene delivery method. However, the treatment failed. Which of the following could be the possible explanations for the failure in the treatment?
- 1 Failure of the expressed protein to be folded into the correct conformation.
 - 2 Integration of the target gene in the enhancer region.
 - 3 Rejection of the vector by the host immune system.
 - 4 The CFTR protein was not expressed in adequate amount.
- A** 1 and 4 only
 - B** 2 and 3 only
 - C** 1, 3 and 4
 - D** 2, 3 and 4

- 40.** Developing fish eggs can be treated to produce a diploid egg. In salmon, such eggs have been fused with haploid salmon sperm to give infertile triploid salmon.

Reproductive organ tissue from diploid trout was transplanted into newly hatched triploid salmon. This tissue matured as the fish grew and the salmon successfully produced viable trout sperm or eggs which resulted in young trout.

Which fish(es) could be seen as genetically modified?

- 1 Salmon providing eggs for treatment
 - 2 Trout providing reproductive organ
 - 3 Young triploid salmon
 - 4 Young trout
-
- A** 3 only
 - B** 1 and 3 only
 - C** 3 and 4 only
 - D** All of the above